

Dot-product logits in GPT: the mathematically precise sentence

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Clean restatement

Yes. Exactly. That sentence is mathematically precise.

I'll restate it cleanly, then justify it from first principles.

Core equation (one line)

For token (i):

$$\text{logit}_i = h^\top E_i$$

Where:

- (h) = compressed context + analysis (final hidden vector)
- (E_i) = representation (embedding) of token (i)

This is a **dot product similarity**.

Why this works (science)

1) What (h) really is

(h) is **not** a summary sentence.

It is a **direction in embedding space** that encodes:

- semantics
- syntax
- positional constraints
- long-range dependencies

It answers:

What kind of token should come next?

2) What (E_i) really is

Each token embedding (E_i) is also a direction, learned so that:

- tokens used in similar contexts point similarly
 - grammar and meaning are embedded geometrically
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3) Dot product meaning

Dot product measures:

- alignment
- agreement
- compatibility

Dot product value	Interpretation
Large positive	Token fits context
Near zero	Token is irrelevant
Negative	Token contradicts context

Then **softmax** turns these compatibilities into probabilities.

Why “compressed” is the right word

The entire history:

$$(x_1, x_2, \dots, x_T)$$

is mapped to:

$$h \in \mathbb{R}^{500}$$

This mapping is:

- lossy
- task-oriented
- predictive (not reconstructive)

Only information useful for predicting the next token survives.

Important nuance

This is **not** cosine similarity:

- norms matter
- magnitude encodes confidence-like information

Temperature (and other sampling controls) later rescales this.

One-line summary

GPT predicts the next token by choosing the token whose embedding has the highest dot-product alignment with the compressed, causally accumulated context vector.

That’s the model in one sentence.

Optional next topics

- why this equals nearest-neighbor under constraints
- how attention learns to shape (h)
- why magnitude is not literally confidence, but often correlates
- why this beats symbolic rules